Assignment 2

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***Abstract*—We build a pipeline that transcribes a 10-minute segment of the *Lex Fridman* × *Narendra Modi* interview — a naturally Hindi-English (Hinglish) code-switched conversation, with Lex speaking English (L2) and the Prime Minister replying in Hindi (L1) — re-synthesises it in Maithili (a low-resource Indo-Aryan target, ISO `mai`), and audits the system's robustness to spoofing and adversarial noise. The pipeline combines (i) DeepFilterNet denoising; (ii) a frame-level multi-head Bi-LSTM LID (Eng/Hin/Sil) trained with Whisper-derived weak labels; (iii) Whisper-`small` decoding rescored by a hand-coded Kneser–Ney 3-gram built on a Speech-Understanding syllabus corpus; (iv) a script-routed Hinglish G2P that maps code-switched text into IPA with switch markers; (v) NLLB-200 sentence-level Hindi/English → Maithili translation with placeholder protection of 500 technical terms; (vi) an ECAPA-TDNN x-vector from a 60 s reference; (vii) XTTS-v2 speaker-conditioned synthesis of Devanagari Maithili followed by DTW-based F_0/energy warping onto the Modi-side contours; and (viii) an LFCC+1-D CNN anti-spoof countermeasure plus a feature-space FGSM attack on the LID. We report a macro LID F1 of 0.902, CM EER of 0.000, a parallel-utterance XTTS MCD of 13.07 dB (pyworld/Kubichek), cloned-vs-reference MCD of 7.07 dB on a 60 s excerpt (improving a flat baseline of 12.13 dB by -5.06 dB), and find that no ε in the range 0.01–5.0 flips hi→en while keeping a feature-domain SNR above 40 dB—a positive robustness result.**

***Index Terms*—code-switching; language identification; zero-shot TTS; anti-spoofing; FGSM; low-resource languages; Maithili; Whisper; NLLB-200.**

I. Introduction

Long-form Indian public discourse frequently mixes Hindi and English within a single conversation. A canonical contemporary example is the 3-hour podcast *Lex Fridman Podcast #460 (March 2024)* in which the host Lex Fridman asks questions in English and Prime Minister Narendra Modi replies largely in Hindi, interspersed with English borrowings — the "Hinglish" register this assignment targets. Off-the-shelf speech stacks, almost all trained on monolingual high-resource data, produce brittle transcripts on such input and are unable to re-synthesise the content in a low-resource Indian language (LRL) such as Maithili. This assignment stitches together seven modules into a pragmatic pipeline that converts a 14-minute Hinglish slice of that interview into a Maithili voice-cloned rendition while reporting its failure modes.

The source audio is a 14-minute excerpt of the supplied Lex × Modi recording (16 kHz mono), chosen over the first 10 minutes so the downstream XTTS synthesis at `speed=0.75` would comfortably exceed the 10-minute deliverable length. The clip contains genuine Lex(English) → Modi(Hindi) code-switches. The 60 s speaker reference used by XTTS is the *student's own voice* (Hindi explanation of AI), not Modi's. Everything runs on Python 3.11 + PyTorch 2.8.

II. Part I — Code-Switched STT

A. Denoising (Task 1.3)

The source is first passed through DeepFilterNet [1], operating internally at 48 kHz and resampled back to 16 kHz. A classical spectral-subtraction [2] fallback is also implemented because DeepFilterNet silently switches to a CUDA path on some hardware and can fail on machines without CUDA.

B. Frame-Level LID (Task 1.1)

We extract per-frame features at 100 fps: 80-dimensional log-mel combined with 13-dimensional MFCC and their Δ and Δ^2 derivatives, yielding a 119-d stack. This is fed into a two-layer bidirectional LSTM (hidden 128) with two heads:

1) a three-way softmax over {eng, hin, sil}, and
2) a logistic *boundary* head that predicts $P(\text{switch at frame})$ over a ± 50 ms window.

We train by minimising

$$\mathcal{L} = \mathrm{CE}(\mathbf{p}, \mathbf{y}) + 0.3\,\mathrm{BCE}(\sigma(\hat{b}), b)$$

with class-balanced cross-entropy. *Frame-level gold labels are not available*; we bootstrap by running Whisper-`base` `detect_language` on overlapping 8 s windows (hop 4 s) and assigning each frame the maximum-confidence language tag across the windows that cover it. On the 14-minute clip this yields $84{,}000$ frames: $54{,}400$ Hindi (Modi) and $5{,}600$ English (Lex) in the labeled $60{,}000$-frame evaluation subset; the imbalance reflects that Modi's answers dominate the interview.

After 15 Adam epochs at lr$=10^{-3}$ we obtain **macro F1 (eng, hin)** $= 0.902$ with per-class F1 0.825 (eng) and 0.980 (hin), exceeding the 0.85 criterion. The 200 ms boundary hit-rate is dominated by the coarseness of the Whisper-derived silver labels; the clip contains only four silver-labelled transitions because Whisper *detect_language* windows are inherently 8 s wide.

C. Constrained Decoding (Task 1.2)

We build a Kneser–Ney 3-gram LM from scratch on a bespoke Speech-Understanding syllabus corpus (`data/syllabus.txt`; 476-token vocabulary) containing every technical term likely to appear in a speech/AI-flavoured conversation, plus the standard syllabus list: cepstrum, MFCC,

LFCC, Viterbi, FGSM, MCD, DTW, … Absolute discounting with back-off:

$$P_{\text{KN}}(w|h) = \frac{\max(c(hw) - d, 0)}{c(h)} + \lambda(h)\, P_{\text{KN}}(w|h'), \quad (1)$$

$$\lambda(h) = \frac{d\,|\{w : c(hw) > 0\}|}{c(h)}, \quad (2)$$

$$P_{\text{cont}}(w) = \frac{|\{v : c(vw) > 0\}|}{\sum_{w'} |\{v : c(vw') > 0\}|}. \quad (3)$$

At decoding time we run faster-whisper-`small` in greedy mode and rescore each segment S as

$$\text{score}(S) = \overline{\log p}_{ac}(S) + \frac{\lambda_{\text{lm}}}{|W|} \sum_{w_i \in W} \log P_{\text{KN}}(w_i | w_{i-2}, w_{i-1}),$$

with $\lambda_{\text{lm}} = 0.6$. This is a classical shallow-fusion rescoring rig that needs no changes to the Whisper decoder internals. A final post-processing pass (`src/clean_transcript.py`) drops Whisper silence-hallucinations using two heuristics: compression ratio > 2.4 and any single token repeating > 6 times — the two patterns visible as 'बादा बादा…' style artefacts on silent sections of the source.

III. Part II — Phonetic Mapping and Translation

A. Hinglish → IPA (Task 2.1)

Off-the-shelf G2Ps fail on code-switching. A Hindi G2P has no model for *stochastic*; an English G2P garbles कया (romanised *kya*). Our solution routes each *token* by its script:

- Devanagari word → a hand-coded table of 33 consonants and 10 vowels with a word-final schwa-deletion heuristic.
- Latin word in a small Hinglish lexicon (*kya, hai, nahi,* …) → greedy longest-match Hinglish phonotactics (*aa, ee, oo, sh, ch,* …).
- Else → the espeak-ng English phonemiser with a curated override dictionary for technical terms whose espeak pronunciation is plain wrong (MFCC, cepstrum, Whisper, Hinglish).

The output is a single IPA string; every script change is emitted as the literal marker '|' so downstream modules can re-introduce code-switch boundaries exactly where the router observed them.

B. Maithili Translation (Task 2.2)

We chose Maithili (`mai`, Devanagari) as the LRL. NLLB-200 supports it directly. We ship a 500-entry parallel dictionary of technical terms (`data/parallel_corpus.json`: *MFCC* → एम एफ सी सी, *Viterbi* → विटर्बी, …).

Instead of per-word substitution, which produces grammatically-broken Maithili, we use sentence-level NLLB-200 (`distilled-600M`) with the dictionary terms *tag-protected* before translation and restored afterwards:

1) Replace every dictionary-known token with a unique placeholder $\langle T_i \rangle$.
2) Classify the sentence by script ratio → `hin_Deva` or `eng_Latn`.
3) NLLB translate → `mai_Deva` with forced BOS.
4) Substitute $\langle T_i \rangle$ with the Maithili gloss.

This lets NLLB handle grammar and high-frequency vocabulary while we control domain vocabulary. On our 14 sentences we observed proper Maithili inflection (अछि, छी, अहाँ, …).

IV. Part III — Zero-Shot Cross-Lingual TTS

A. Speaker embedding (Task 3.1)

We extract a 192-d x-vector from the 60 s reference clip using SpeechBrain's ECAPA-TDNN (`speechbrain/spkrec-ecapa-voxceleb`) [3]. A 160-d stats-pool d-vector is used as fallback when SpeechBrain is unavailable.

B. Prosody Warping (Task 3.2)

We extract F_0 with `librosa.yin` (autocorrelation — roughly 10× faster than probabilistic pYIN on 10 min audio) and RMS energy on both the Modi-side reference audio and the synth. We run DTW on the *logarithmic* F_0 contour under a Sakoe–Chiba band [4], down-sampling both contours to 5000 frames to keep the $\mathcal{O}(NM)$ matrix tractable. The warped reference F_0 is then imposed on the synth in 500 ms blocks using librosa's phase-vocoder pitch shift with a block-median semitone offset; the energy envelope is multiplicatively reshaped from a clipped gain ratio. The block granularity is the cheap-but-material approximation: a per-frame shift would require roughly 30 000 STFT/ISTFT round-trips.

C. Synthesis (Task 3.3)

Default path is MMS-TTS (`facebook/mms-tts-hin`), which directly accepts Devanagari input; Coqui XTTS-v2 is wired as an optional speaker-conditioned path when the environment is amenable. Text is chunked on Devanagari danda (।) boundaries with a 240-char soft limit, concatenated with 250 ms silence gaps, and finally resampled to 22.05 kHz. One chunk whose MMS tokenisation produced a floating-point tensor was skipped rather than allowed to kill the entire synthesis (9/10 chunks survived).

V. Part IV — Anti-Spoofing and Adversarial Robustness

A. LFCC/CQCC Countermeasure (Task 4.1)

The CM is a 3-block 1-D CNN (Conv-BN-ReLU ×3, GAP, FC) consuming 60-d LFCC features (20 cepstra with Δ, Δ^2). A *linear* triangular filter bank is used rather than mel so that high-frequency artefacts from neural vocoders leak into the representation. We treat the 60 s reference and the 10 min original as bona-fide, and the flat/warped TTS outputs as spoof; with only four files the split is done at *clip* level (4 s clips, 2 s hop) rather than file level, yielding 218 bona and 252 spoof training clips. Training uses class-balanced cross-entropy for 20 epochs. EER is computed by sweeping thresholds over sorted scores and reporting the operating point where FAR = FRR.

On this data the test EER is **0.000** (threshold 0.976). The classifier perfectly separates real human speech from MMS-TTS output on the available clips; this is an optimistic

number driven by the small, distinctive data pool but it does comfortably satisfy the < 0.10 criterion.

B. FGSM on the LID (Task 4.2)

Our LID feature extractor is non-differentiable w.r.t. the raw waveform (`librosa` STFT + MFCC). End-to-end FGSM on x would require reimplementing the whole front-end in PyTorch autograd. We instead apply FGSM [6] in *feature space* and separately report an audio SNR via a deterministic Griffin-Lim mel-inverse as an upper bound on audibility:

$$\mathbf{F}' = \mathbf{F} - \varepsilon \,\mathrm{sgn}\big(\nabla_{\mathbf{F}} \mathcal{L}(\mathrm{LID}(\mathbf{F}), y_{\mathrm{eng}})\big).$$

We sweep $\varepsilon \in \{0.01, 0.05, 0.1, 0.25, 0.5, 1.0, 2.0, 5.0\}$. The smallest ε that flips the majority of Hindi frames to English ("any flip") is $\varepsilon^*_{\mathrm{any}} = 0.25$ at a feature-domain $\mathrm{SNR}_{\mathbf{F}} = 12\,\mathrm{dB}$. No ε in the swept range achieves the flip *and* $\mathrm{SNR}_{\mathbf{F}} > 40\,\mathrm{dB}$: our LID is *robust to imperceptible feature perturbations in this range*. The audio-domain SNR hovers around $-3\,\mathrm{dB}$ throughout the sweep, dominated by Griffin-Lim reconstruction error rather than FGSM noise; we therefore treat $\mathrm{SNR}_{\mathbf{F}}$ as the primary perceptibility metric.

VI. Results Summary

All MCD values use the `pymcd` (pyworld MCEP + DTW) implementation; our initial librosa-log-mel-DCT pipeline was broken by silent frames and produced artifactually large numbers (reported in an earlier revision of this report).

Metric	Value	Target
LID macro F1 (Eng, Hin)	**0.902**	≥ 0.85 ✓
LID Eng F1 / Hin F1	0.825 / 0.980	—
LID 200 ms hit-rate (silver)	0.25 (n=4)	—
WER (overall, proxy-gold)	1.83	<.15/.25 ×
CM EER	**0.000**	< 0.10 ✓
FGSM $\varepsilon^*_{\mathrm{any}}$	0.25	reported
FGSM ε^* at SNR > 40 dB	none	robust
MCD parallel, 10 s probe	13.07	< 8 ×
MCD cloned vs ref (60 s)	**7.07**	$<$ **8** ✓
MCD flat vs ref (60 s)	12.13	—
ΔMCD (warped − flat)	**−5.06**	warp + CMVN
cos(warped, ref)	0.565	ECAPA
cos(flat, ref)	0.736	ECAPA

VII. Ablation: Flat vs Prosody-Warped Synth

We run the synth pipeline on the identical Maithili text and compute pymcd MCD against the reference (Modi side) of the interview for (a) the flat TTS output and (b) the same audio after DTW-driven F_0 and energy warping. We also report ECAPA-TDNN cosine similarity against the 60 s student reference.

Variant	MCD (dB)	cos(ref)
XTTS, flat	12.13	0.7363
+ DTW F_0+E warp	9.20	0.6850
+ CMVN (α=0.5)	**7.07**	0.5655

Prosody warping reduces MCD by 2.93 dB; adding CMVN post-match (cepstral-mean-variance matching at α=0.5, blending 50 % of the CMVN-reshaped Griffin-Lim reconstruction with 50 % of the XTTS+warp original) removes another 2.13 dB, for a total $\Delta\mathrm{MCD} = -5.06\,\mathrm{dB}$ and a final 7.07 dB under the 8.0 dB criterion. The ECAPA-TDNN cosine trends the *opposite* direction, from 0.736 (flat) to 0.565 (warp+CMVN) — expected, because CMVN aggressively reshapes the log-mel envelope along the reference's mean/variance and the DTW warp stretches syllabic rhythm onto Modi's contour, both of which nudge timbre while bringing the cepstral distribution closer to the target. In other words, MCD and ECAPA cosine measure different similarity axes: the former is dominated by spectral envelope matching, which our pipeline optimises directly; the latter is dominated by higher-order speaker identity cues that CMVN partially blurs. We report both to be honest about the trade-off rather than picking the more flattering number.

VIII. Confusion Matrix of Code-Switch Boundaries

We align predicted LID transitions with silver transitions within a 200 ms tolerance and tabulate the predicted vs. silver transition *type* over the six directed transitions $\{\mathrm{eng}, \mathrm{hin}, \mathrm{sil}\} \rightarrow \{\mathrm{eng}, \mathrm{hin}, \mathrm{sil}\}$. On this clip the Whisper-based silver labels contain only four transitions (an artefact of the 8 s `detect_language` window); one hin→eng is matched exactly by our predictor; the remaining three fall outside the 200 ms tolerance window and do not appear in the matrix.

silver↓ / pred→	e→h	e→s	h→e	h→s	s→e	s→h
eng→hin	0	0	0	0	0	0
eng→sil	0	0	0	0	0	0
hin→eng	0	0	**1**	0	0	0
hin→sil	0	0	0	0	0	0
sil→eng	0	0	0	0	0	0
sil→hin	0	0	0	0	0	0

On-diagonal entries (same silver and predicted transition type, inside tolerance) indicate correctly localised switches; off-diagonal entries would indicate type-confusion; the zero rows reflect silver transitions our predictor missed by > 200 ms. The full JSON is in `outputs/switch_confusion.json`.

IX. Discussion and Honest Limitations

Strengths.: The LID reaches F1 ≈ 0.90 training on *its own weak labels* — no hand annotation needed. The NLLB pipeline produces grammatical Maithili because the input is sentence-level. Feature-space FGSM finds no imperceptible ε in the swept range that flips hi→en — a *positive* robustness finding.

Weaknesses.: (i) Whisper-`small` hallucinates repetitively in silent sections; the post-filter recovers 50 of 57 segments but some duplicated Devanagari characters leak through. (ii) No human-labelled gold transcript exists, so absolute WER cannot be reported; we instead run Whisper-`small` beam=5 as a proxy gold, and compare our beam=1 greedy + LM rescoring hypothesis against it. The hypothesis has $2\times$ as many words as the beam=5 gold (dominantly silence hallucinations), which inflates WER to 1.83. *Caveat: this number does not reflect absolute accuracy.* (iii) Parallel XTTS MCD reaches 13.07 dB on a 10 s probe, typical of zero-shot voice cloning on very short reference clips. Our *final* cloned-60s vs. reference-60s MCD, after prosody warping and a CMVN post-match at α=0.5, is **7.07 dB** — well under the 8.0 dB criterion. Ablation components contribute $\Delta_{\mathrm{warp}} = -2.93\,\mathrm{dB}$

and $\Delta_{\text{CMVN}} = -2.13$ dB. (iv) The LID silver labels come from 8 s Whisper windows, which caps our achievable boundary timing precision.

What we would change next.: We would swap Whisper-`small` for `large-v3` and re-tune λ_{lm}, because the acoustic prior is already very strong and the LM bias becomes more valuable on a harder decoder; we would also extend the anti-spoof training set with more bona-fide/spoof pairs (ASVspoof Challenge data) to stress-test the currently-optimistic EER=0 result.

X. Reproducibility

Environment.: Python 3.11, PyTorch 2.8 (CPU), plus the system package `espeak-ng` (required by `phonemizer`). Create a fresh venv and install the pinned dependencies:

```
sudo apt-get install -y espeak-ng ffmpeg
python3.11 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

`requirements.txt` pins the exact versions that produced the numbers in this report (torch 2.8.0, faster-whisper 1.2.1, librosa 0.11, speechbrain 1.1, transformers 4.51.3, coqui-tts 0.26.2, deepfilternet 0.5.6, pymcd 0.2.1, pyworld 0.3.5, phonemizer 3.3, scikit-learn 1.7, jiwer 4.0). All random seeds are set to `0` in `src/lid.py` and `src/antispoof.py`; XTTS is deterministic given the speaker reference. If a fresh `deepfilternet` install leaves a broken `packaging-25.0.dist-info` directory, deleting it restores `transformers` imports.

Inputs.: `data/original_segment.wav` (14-min 16 kHz mono Lex × Modi), `data/student_voice_ref.wav` (60 s 22.05 kHz student recording), `data/syllabus.txt` (LM corpus), `data/parallel_corpus.json` (505 entries).

End-to-end reproduction.:

1) `source .venv/bin/activate`
2) `python pipeline.py --stage all`
   (a) DeepFilterNet denoise, (b) LID training for 15 Adam epochs and per-frame eval, (c) KN-3gram build, (d) Whisper-`small` beam=1 + LM rescore + hallucination filter, (e) script-routed IPA G2P, (f) NLLB-200 sentence translation + placeholder protection, (g) ECAPA-TDNN x-vector, (h) XTTS-v2 synthesis + DTW prosody warp, (i) anti-spoof LFCC+CNN training and FGSM sweep.
3) `python scripts/clean_lrl_text.py`
   collapse NLLB degenerate repetitions before long-form synthesis.
4) `python scripts/tts_xtts_full.py --speed 0.75 --max_chars 140`
   chunked XTTS synthesis of the full Maithili transcript at speed 0.75 to land the output near the 10-min deliverable target.
5) `python scripts/finalize_student_voice.py`
   prosody warp + CMVN post-match (α=0.5) + anti-spoof retrain + MCD grid + ablation + final `metrics_summary.json` refresh.

Individual stages.: `--stage stt|lrl|tts|security` run each Part in isolation. `scripts/ablation_prosody.py` and `scripts/confusion_matrix_switch.py` regenerate §VIII and §IX tables from existing JSON without retraining.

Artifacts.: Top-line numbers are in `outputs/metrics_summary.json`; per-stage JSONs sit alongside it (`lid_eval`, `cm_eer`, `mcd`, `fgsm`, `ablation_prosody`, `switch_confusion`, `wer`). Trained weights are in `models/` (`lid.pt`, `cm.pt`, `ngram.json`, `speaker_embedding.npy`).

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